

News Agent

Daily AI-powered news aggregator for the Meridian platform

Overview

The News Agent is a fully autonomous pipeline that runs twice daily (06:00 and 18:00 UTC) to collect, curate, summarise, and publish news from 20+ RSS feeds across four geographic regions. It uses Gemini 2.5 Flash for intelligent article curation and Gemini image models to enhance every thumbnail before storing it in the Meridian media service.

Schedule & Invocation

Cron schedule	0 6,18 * * * (06:00 and 18:00 UTC daily)
Trigger	GitHub Actions workflow or manual: <code>python -m meridian_agents.news_agent</code>
Entry point	<code>meridian_agents/news_agent/main.py</code> → <code>run_news_agent()</code>
SDK	openai-agents (Runner + Agent)
Primary model	gemini-2.5-flash (via Google Generative Language OpenAI-compatible endpoint)
Fallback model	None — GEMINI_API_KEY is required
Image model	gemini-3.1-flash-image (fallbacks: gemini-2.5-flash-image, gemini-3-pro-image)

Agent Flow

The agent runs as a single-agent loop with one tool (`save_news`). All heavy pre-processing happens before the agent is invoked:

Step	Description
1. RSS Fetch (pre-agent)	<code>fetch_region_news()</code> is called for all 4 regions in parallel using <code>ThreadPoolExecutor</code> . Each region queries its configured feeds via <code>feedparser</code> .
2. Dedup & Sort (pre-agent)	Articles are deduplicated by normalised title and sorted newest-first. A 36-hour freshness window is applied; falls back to all articles if <3 fresh ones exist.

Step	Description
3. Agent Curation	Gemini 2.5 Flash receives the full article batch as JSON and is instructed to select exactly 10 stories (3 world, 3 USA, 2 India, 2 Odisha), write ~100-word summaries, and call <code>save_news()</code> .
4. OG Image Enrichment	For items without an <code>imageUrl</code> , <code>_enrich_images()</code> scrapes the article's <code>og:image</code> in parallel (<code>ThreadPoolExecutor</code>).
5. Gemini Image Enhancement	<code>_enhance_all_images()</code> downloads every thumbnail, sends it to <code>gemini-3.1-flash-image</code> with a sharpening prompt, and uploads the enhanced version to the media service.
6. Publish	<code>publisher.save_news_items()</code> posts all 10 items to <code>POST /api/news/refresh</code> , replacing the previous news batch atomically.

News Sources (RSS Feeds)

Region	Sources
World	BBC World, CNN Edition, The Guardian World, Al Jazeera
USA	BBC US & Canada, CNN US, NPR
India	NDTV, Times of India, The Hindu National
Odisha	OTV, Pragativadi, The Hindu Odisha

Tools

Tool / Helper	Purpose
<code>fetch_region_news(region)</code>	Aggregates articles from all RSS feeds in a region using parallel fetching. Returns up to 10 deduplicated, sorted articles.
<code>save_news(items_json)</code>	@function_tool called by the LLM agent. Deserialises JSON, runs enrichment and enhancement pipelines, then publishes via the Meridian API.
<code>_enrich_images(items)</code>	Internal helper: fetches <code>og:image</code> / <code>twitter:image</code> for articles missing a thumbnail.
<code>_gemini_enhance(bytes, mime)</code>	Internal: sends image to Gemini image model with sharpening prompt; returns <code>(enhanced_bytes, mime)</code> or <code>None</code> .
<code>_enhance_all_images(items)</code>	Internal orchestrator: runs enhancement for all items in parallel (max 5 workers).

Tool / Helper	Purpose
<code>_upload_to_media(buf, mime, alt)</code>	Internal: uploads bytes to <code>/api/media/upload</code> with agent JWT; returns URL.
<code>_download_image(url)</code>	Internal: downloads image URL with browser-like headers; returns (bytes, mime).

Models Used

Model	Role
<code>gemini-2.5-flash</code>	Curation & summarisation agent — selects 10 articles, writes summaries, calls <code>save_news()</code>
<code>gemini-3.1-flash-image</code>	Primary image enhancement model (sends image + sharpening prompt, returns enhanced image)
<code>gemini-2.5-flash-image</code>	Fallback image model if primary is unavailable
<code>gemini-3-pro-image</code>	Second fallback image model

Authentication

`make_agent_jwt(name='News Agent', email='news-agent@meridian.internal')` from `meridian_agents/auth.py` produces a 1-hour HS256 JWT signed with the platform secret (`myblogs-secret-key-2024`). The JWT carries `role=admin` and is used for all API calls to the Meridian platform.

Key Files

File	Purpose
<code>news_agent/main.py</code>	Entry point, agent build, RSS pre-fetch, asyncio runner
<code>news_agent/tools.py</code>	All tool functions: <code>fetch</code> , <code>enrich</code> , <code>enhance</code> , <code>upload</code> , <code>save_news</code>
<code>news_agent/publisher.py</code>	<code>POST /api/news/refresh</code> — replaces live news batch
<code>meridian_agents/auth.py</code>	Shared JWT helper used for authenticated API calls